KPI Autopsy | Banking metrics

Intake 2016-01-01 to 2017-01-01 (end exclusive) | Observed through 2017-02-01T14:11:03.499000Z | UTC

Applications with an accepted offer

54.68%

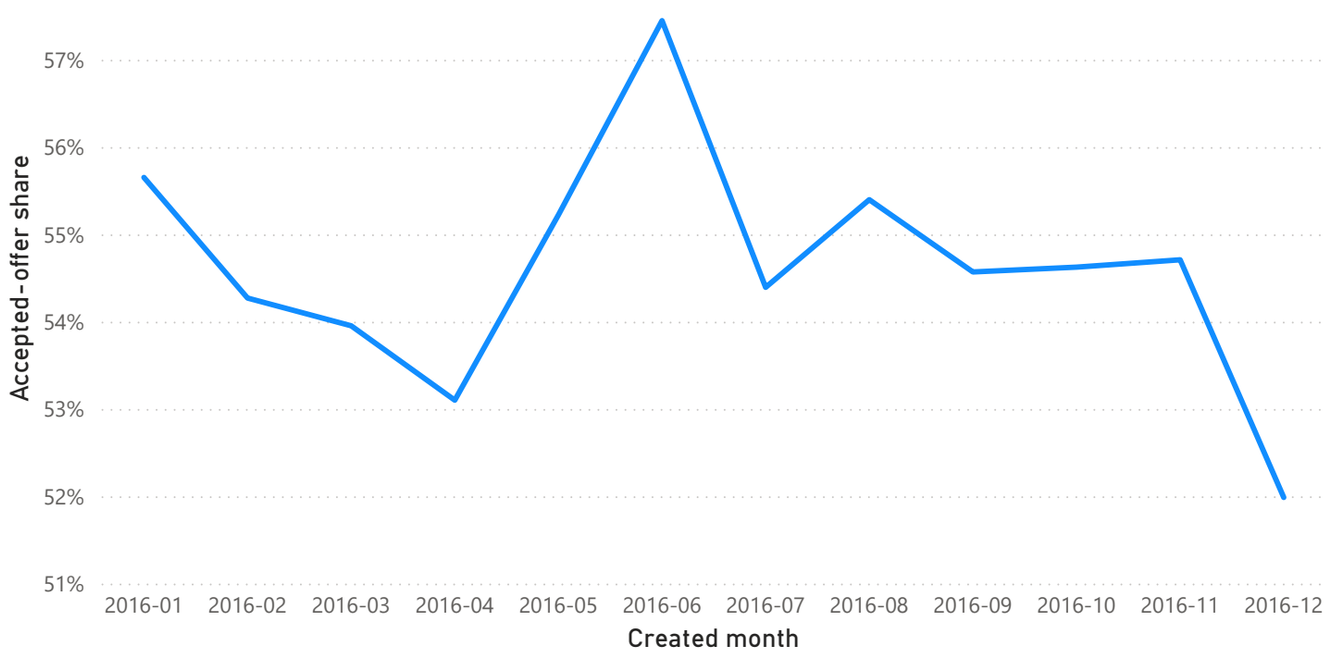
Mean time to first offer (calendar hours)

37.39

Open applications under the declared state rule

98

Accepted-offer share by intake month



Compare loan purposes

Loan goal	Applications	Accepted-offer share
Boat	201	57.21%
Business goal	30	26.67%
Car	9,328	51.32%
Caravan / Camper	369	56.91%
Debt restructuring	2	0.00%
Existing loan takeover	5,601	54.88%
Extra spending limit	625	52.96%
Home improvement	7,669	58.56%
Motorcycle	275	50.18%
Not speficied	1,065	41.22%
Other, see explanation	2,985	50.59%
Remaining debt home	842	64.61%
Tax payments	152	47.37%
Unknown	2,365	63.81%
Total	**31,509**	**54.68%**

Definitions matter

Accepted offer is an observed event outcome. Turnaround covers offered applications; backlog follows a declared event-state rule.

Source: Boudewijn van Dongen (2017), BPI Challenge 2017, 4TU.ResearchData. doi:10.4121/uuid:5f3067df-f10b-45da-b98b-86ae4c7a310b

Investigate a misleading report

Fixed export cohort. Select a report row to filter its evidence; the reference SQL is saved beside the audit JSON.

Reported and reconciled values — read the unit column

Title	Status	Reported value	Reference value	Difference	Unit	Affected applications
Accepted at the wrong stage	mismatch	100.00	54.68	45.32	percent	14,281
Accepted-only turnaround population	mismatch	33.34	37.39	-4.05	hours	14,281
Clean accepted-offer report	reconciled	54.68	54.68	0.00	percent	0
Clean backlog report	reconciled	98.00	98.00	0.00	applications	0
Clean first-offer report	reconciled	37.39	37.39	0.00	hours	0
Offer join changes application grain	mismatch	56.16	54.68	1.49	percent	8,559
Offer-date cohort replaces intake cohort	mismatch	54.73	54.68	0.05	percent	110

Affected applications and their contributions

Report id	Application id	Reason	Reported rows	Reference rows	Reported numerator	Reference numerator	Offer count
offer-cohort	Application_100853745	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1067190799	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1090720403	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1107993134	missing application	0	1	0.00	1.00	1
offer-cohort	Application_1120379399	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1168372021	missing application	0	1	0.00	1.00	1
offer-cohort	Application_117342811	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1183810144	missing application	0	1	0.00	0.00	1
offer-cohort	Application_1217378577	missing application	0	1	0.00	1.00	2
offer-cohort	Application_1217715016	missing application	0	1	0.00	0.00	2
offer-cohort	Application_1252751917	missing application	0	1	0.00	0.00	2
offer-cohort	Application_1281199569	missing application	0	1	0.00	1.00	1

Read the gap in the right unit

Rate differences are percentage points. Other differences keep the metric's unit. Use an application ID from the table to look up its events on page 3.

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Follow an application to its source events

Select an application, then inspect its timestamps, lifecycle and offer IDs. Events are bounded by the export cutoff.

Application ID

Application id

Search

- Application_1000086665
- Application_1000158214
- Application_1000311556
- Application_1000334415
- Application_1000339879
- Application_100034150
- Application_1000386745
- Application_1000474975
- Application_1000557783
- Application_1000604502
- Application_1000610355
- Application_1000647090
- Application_1000671285
- Application_1000691650
- Application_1000806256
- Application_1000856802
- Application_1000867366

Source event history

Application id	Occurred at	Event row	Activity	Lifecycle	Offer id	Source event id
Application_652823628	2016-01-01T09:51:15.304000Z	1	A_Create Application	complete		Application_652823628
Application_652823628	2016-01-01T09:51:15.352000Z	2	A_Submitted	complete		ApplState_1582051990
Application_652823628	2016-01-01T09:51:15.774000Z	3	W_Handle leads	schedule		Workitem_1298499574
Application_652823628	2016-01-01T09:52:36.392000Z	4	W_Handle leads	withdraw		Workitem_1673366067
Application_652823628	2016-01-01T09:52:36.403000Z	5	W_Complete application	schedule		Workitem_1493664571
Application_652823628	2016-01-01T09:52:36.413000Z	6	A_Concept	complete		ApplState_642383566
Application_1691306052	2016-01-01T10:16:11.500000Z	41	A_Create Application	complete		Application_16913060
Application_1691306052	2016-01-01T10:16:11.549000Z	42	A_Submitted	complete		ApplState_284636842
Application_1691306052	2016-01-01T10:16:11.740000Z	43	W_Handle leads	schedule		Workitem_831373279
Application_1691306052	2016-01-01T10:17:31.573000Z	44	W_Handle leads	withdraw		Workitem_1299098074
Application_1691306052	2016-01-01T10:17:31.584000Z	45	W_Complete application	schedule		Workitem_1703931302
Application_1691306052	2016-01-01T10:17:31.594000Z	46	A_Concept	complete		ApplState_421204851
Application_428409768	2016-01-01T11:19:38.177000Z	72	A_Create Application	complete		Application_428409768
Application_428409768	2016-01-01T11:19:38.235000Z	73	A_Submitted	complete		ApplState_105207793
Application_428409768	2016-01-01T11:19:38.914000Z	74	W_Handle leads	schedule		Workitem_1185806608
Application_428409768	2016-01-01T11:20:37.391000Z	75	W_Handle leads	withdraw		Workitem_835967796
Application_428409768	2016-01-01T11:20:37.409000Z	76	W_Complete application	schedule		Workitem_755889823
Application_428409768	2016-01-01T11:20:37.422000Z	77	A_Concept	complete		ApplState_1727723699
Application_1746793196	2016-01-01T12:34:53.911000Z	117	A_Create Application	complete		Application_174679319
Application_1746793196	2016-01-01T12:34:53.950000Z	118	A_Submitted	complete		ApplState_282472804
Application_1746793196	2016-01-01T12:34:54.320000Z	119	W_Handle leads	schedule		Workitem_1676837235
Application_828200680	2016-01-01T13:00:04.360000Z	201	A_Create Application	complete		Application_828200680
Application_828200680	2016-01-01T13:00:04.398000Z	202	A_Submitted	complete		ApplState_420123528

A row can be traced

event_row is the stable position in the pinned XES file; source_event_id and the original timezone-bearing timestamp are retained in the export.

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Evaluate the documented fault families

Separate September–December cohorts; original and equivalent wrapped SQL; clean controls included. No model is trained.

Evaluation scenarios

56

Detection recall on exercised known faults

100.0%

False alarms on clean reports

0

Every evaluation case remains inspectable

Month	Scenario	Variant	Expected issue	Detected issues	Exercised	Passed	Query ms
2016-09	clean-backlog	original	NONE	NONE	1	1	11.96
2016-09	clean-backlog	wrapped	NONE	NONE	1	1	11.22
2016-09	clean-delay	original	NONE	NONE	1	1	12.12
2016-09	clean-delay	wrapped	NONE	NONE	1	1	11.64
2016-09	clean-rate	original	NONE	NONE	1	1	11.45
2016-09	clean-rate	wrapped	NONE	NONE	1	1	11.26
2016-09	offer-cohort	original	COHORT_MISMATCH	COHORT_MISMATCH	1	1	7.89
2016-09	offer-cohort	wrapped	COHORT_MISMATCH	COHORT_MISMATCH	1	1	8.59
2016-09	offer-join	original	GRAIN_MISMATCH	GRAIN_MISMATCH	1	1	20.49
2016-09	offer-join	wrapped	GRAIN_MISMATCH	GRAIN_MISMATCH	1	1	20.73
2016-09	outcome-filter	original	OUTCOME_FILTER	OUTCOME_FILTER	1	1	10.53
2016-09	outcome-filter	wrapped	OUTCOME_FILTER	OUTCOME_FILTER	1	1	10.12
2016-09	wrong-stage	original	EVENT_DEFINITION_MISMATCH	EVENT_DEFINITION_MISMATCH	1	1	10.65
2016-09	wrong-stage	wrapped	EVENT_DEFINITION_MISMATCH	EVENT_DEFINITION_MISMATCH	1	1	10.83
2016-10	clean-backlog	original	NONE	NONE	1	1	12.23
2016-10	clean-backlog	wrapped	NONE	NONE	1	1	13.17
2016-10	clean-delay	original	NONE	NONE	1	1	11.39

Read the scope with the score

Unexercised faults remain visible. Scores cover known error families and separate cohorts, not arbitrary SQL or unseen business rules.

Source: Boudewijn van Dongen (2017), BPI Challenge 2017, 4TU.ResearchData. doi:10.4121/uuid:5f3067df-f10b-45da-b98b-86ae4c7a310b